

ELEC451: POWER ELECTRONICS**LAB 1: POWER QUALITY AND BASIC UNCONTROLLED RECTIFIERS****OBJECTIVES**

1. Apply concepts of AC/DC line rectification
2. Learn how to implement basic power electronic circuits
3. Get familiar with a power electronics simulation platform
4. Measure THD, PF, and DPF in a rectifier circuit
5. Compare the results of the simulation to those of the experimental platform

LAB REPORT

- Due Date: one week after the lab session (upload to Canvas).
- **Tips:** follow the guidelines provided in Lab 0, particularly the ones about capturing adequate waveforms, to avoid losing marks unnecessarily.

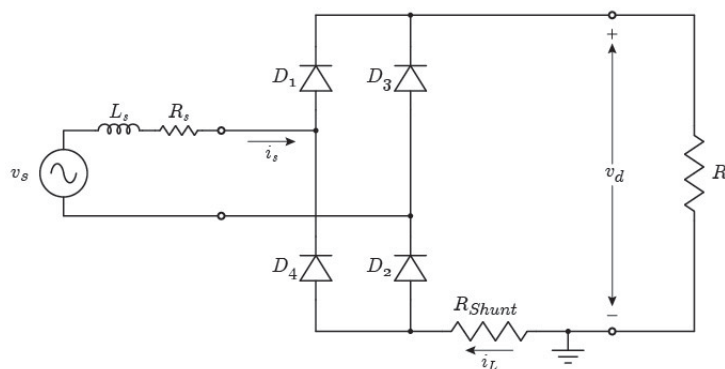
INSTRUCTIONS

- Read and follow each step carefully, especially the experiment part.
- Before turning on the power supply, ask your TA to double check your setup.
- Turn off the power supply before reconfiguring the circuit.

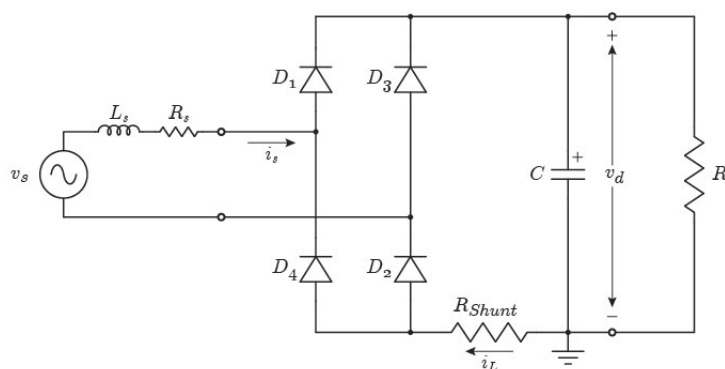
ACTIVITY 1 – SIMULATION

- a. Build the following circuits using PSIM. Create a new model and place the components for each circuit.
- b. Using $V_s = 12V_{rms}$ and $F_s = 60Hz$, simulate, plot and print the input voltage V_s , output voltage V_d , input current i_s and current i_L for each of the cases shown below. Consider $L_s = 1mH$, $R_s = 2\Omega$, $L = 1mH$, $C = 50\mu F$, $R = 500\Omega$ and $R_{shunt} = 1\Omega$.
- c. For each of the cases, calculate the crest factor and the total harmonic distortion (THD) using the equations provided in class. Validate the THD results using the measurement tool provided by PSIM. Make sure to analyze the steady-state waveforms, disregarding transients. If possible, indicate displacement power factor (DPF) and power factor (PF). If not, explain why.
- d. In your report, show the mentioned waveforms and calculations, and make relevant comments. When saving the plots, ensure that you are using the correct simulation time step size. Follow the image formatting guidelines seen in Lab 0.

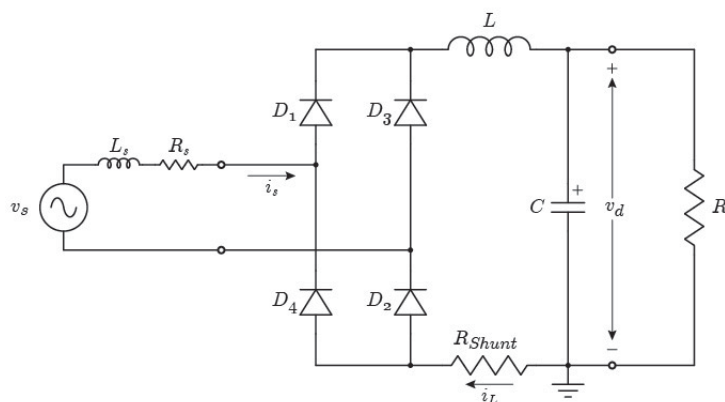
Tips: Remember to check if the magnitude of the FFT is in dB or Amp.



Circuit 1 – Single-phase full bridge rectifier with resistive load



Circuit 2 – Single-phase full bridge rectifier with output capacitive filter and resistive load



Circuit 3 – Single-phase full bridge rectifier with LC filter and resistive load

ACTIVITY 2 – EXPERIMENTAL WORK

- Implement the same circuits described in Activity 1 using the uncontrolled rectifier board that will be provided.
- Using the oscilloscope, obtain the experimental captures of the input voltage V_s ($12V_{rms}$), output voltage V_d , input current i_s and current i_L for each of the cases. The ground and connections should be as indicated in the circuits. The input current must be measured using a current probe, the input voltage using a differential probe, and the output voltage and inductor current (measured at the shunt resistor) employing regular voltage probes.

Important:

- Before turning on the power supply, ask your TA to validate the circuit.
 - The ground connector of the regular voltage probes must be connected to the same point (ground/earth); otherwise, it will cause a short circuit!
 - For taking experimental captures, make sure to follow the guidelines provided in Lab 0.
- For each of the cases, determine the crest factor and the total harmonic distortion (THD). If possible, indicate displacement power factor (DPF) and power factor (PF). If not, explain why. In order to determine the THD, the FFT calculation must be performed. The estimation of the distortion, based on the amplitude of the harmonics, is given by the following equation:

$$THD_i = 100 \frac{\sqrt{i_{2(RMS)}^2 + i_{3(RMS)}^2 + \dots}}{i_{1(RMS)}} \%$$

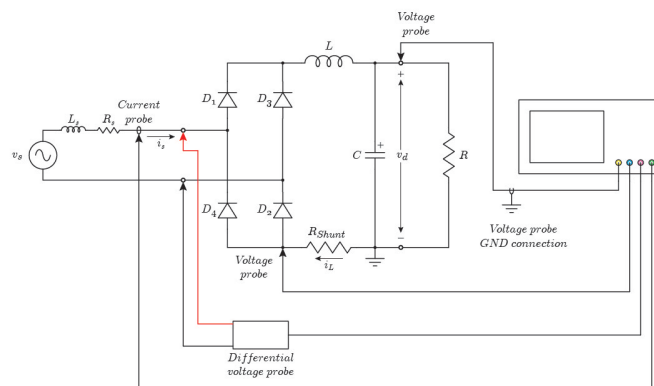
The THD can also be obtained by measuring the RMS value of the input current and its fundamental component.

$$THD_i = 100 \frac{\sqrt{i_{(RMS)}^2 - i_{1(RMS)}^2}}{i_{1(RMS)}} \%$$

Include time-domain and FFT captures, measurements, calculation procedures, final values and any other relevant details in your report. Show the mentioned waveforms and calculations, and make relevant comments.

Tips: Remember to check if the magnitude of the FFT is in dB or Amp.

$$i[dB] = 20 \log(i[A]).$$

**Experimental Setups – Single-phase full bridge rectifier with LC filter and resistive load**